

CS15L Programming Languages

Final Project Instructions

React Native Mobile App Development

Dear Devs,

This is your Final Project for our Programming Languages course. Each group already has an approved project title, and you will now move forward with the development phase.

Your final output will be presented in a Hackathon-style panel presentation. Bring your best work—clean code, modern UI, and working features.

Final Requirements

1. Custom App Logo & Branding

Your team must create and apply its own app branding.

- Create your own **unique app logo**.
- Apply your branding consistently throughout the application.
- Use a consistent:
 - Color palette
 - Icons
 - Spacing
 - Typography
 - Button and component styles

Your application should have a distinct identity and should **not look like a default React Native/Expo template**.

2. Multi-Screen React Native App

Your application must contain **at least 4 functional screens**, including:

- **1 Home/Dashboard screen**
- **At least 2–3 additional functional screens**
- Smooth navigation between screens using **React Navigation**

Your application may use:

- Stack Navigation
- Bottom Tab Navigation
- Drawer Navigation
- Nested Navigation

The navigation structure should be appropriate for your application.

3. Transactional Function

Your application must implement **at least one meaningful transaction**.

The transaction must:

- Involve **user input**
- Allow the user to **add, edit, or save information**
- Update data stored in the database
- Display the updated information in the application

Examples:

- Adding a customer record
- Creating a reservation
- Submitting an order
- Adding a product
- Recording a transaction
- Creating a task
- Registering an appointment
- Saving a list item
- Updating a user profile

The transaction should represent an actual function of your approved project.

4. Database Integration — Choose ONE

Your application must use **one database solution**.

Option A — SQLite Local Database

Use a SQLite database stored locally on the device.

CS15L Programming Languages
Final Project Instructions
React Native Mobile App Development

OR

Option B — API Backend

Use an API/backend such as:

- Node.js
- Laravel
- ASP.NET
- Firebase
- Supabase
- Other REST API

Database Requirement

Your application must demonstrate all four basic database operations:

CREATE → READ → UPDATE → DELETE

Your application must successfully:

- **Save/Create** data
- **Fetch/Read** data
- **Update/Edit** data
- **Delete** data

Important: AsyncStorage alone does **not** satisfy the database requirement. Your team must use SQLite or an actual backend/API/database solution.

5. Modern UI / UX

Your application must demonstrate a modern and user-friendly interface.

Requirements:

- Responsive layout
- Proper padding and spacing
- Proper alignment
- Consistent color palette
- Clean typography
- Appropriate icons
- User-friendly interactions

- Clear navigation
- Proper buttons and input fields
- Appropriate loading/error states where necessary
- No default React Native/Expo boilerplate appearance

The application should look like a **real-world mobile application**, not simply a collection of screens.

6. React Native Technical Requirements

Your project must demonstrate proper use of React Native concepts.

Your code should appropriately use:

- React Native components
- Functional components
- `useState`
- `useEffect` where appropriate
- Props
- Event handling
- Lists such as `FlatList`
- Forms and user input
- React Navigation
- Database/API operations
- Appropriate state management

Students should organize their project using a **clear and logical folder structure**.

7. Hackathon-Style Final Presentation

You will present your application **live to a panel**.

Presentation Format

1. App Pitch — 1 minute

Explain:

- What your application does

CS15L Programming Languages
Final Project Instructions
React Native Mobile App Development

- What problem it solves
- Who the target users are

2. Live Demo — 5 minutes

Demonstrate the actual working application.

Your demonstration should include:

- Home screen
- Navigation
- Major features
- User input
- Transaction
- Database operation
- Save/Create
- Fetch/Read
- Update
- Delete

3. Technical Discussion — 2–3 minutes

Explain:

- React Native/Expo
- React Navigation
- Components
- State management
- Database/API
- How your transaction works
- How data flows through your application

4. Q&A from Panel

Be prepared to answer questions about:

- Your code
- Application logic
- Navigation
- Database
- React Native concepts
- Design decisions
- Problems encountered during development

Dress neatly (Smart Attire) and prepare your talking points.

Final schedule: TBD (will be announced soon).

Submission Requirements

Submit the following through **Google Classroom**:

1. APK File

Submit your working Android application:

- Debug APK **or**
- Release APK

The APK must be installable and demonstrate the required features.

2. GitHub Repository Link

Submit the link to your team's GitHub repository.

The repository should contain:

- Complete React Native source code
- Project configuration files
- Required assets
- Database/API-related code
- README/documentation

3. Short Documentation PDF

Your PDF must contain:

- **App Title**
- **Team Members**
- **App Description**
- **Screenshots of Major Screens**
- **Database Structure**
- **Tech Stack Used**

CS15L Programming Languages
Final Project Instructions
React Native Mobile App Development

You may also include:

- System architecture
- Navigation flow
- Database/API flow
- Key features
- Instructions for running the application

CS15L Programming Languages
Final Project Instructions
React Native Mobile App Development

Rubric

Category	Description	Points
A. App Design (UI/UX)	Modern UI, clean layout, responsive design, proper spacing, consistent theme, appropriate React Native components	20 pts
B. Functionality & Features	Core features working, multi-screen navigation, user input, and smooth transactions	25 pts
C. Database Integration	Ability to create/save, fetch/read, update, and delete data using SQLite or an API/backend	20 pts
D. Code Quality	Readability, logical folder structure, reusable components, proper state management, appropriate React Native practices, minimal bugs	15 pts
E. Presentation (Hackathon Style)	Clarity, confidence, demonstration flow, technical explanation, and ability to answer panel questions	10 pts
F. Documentation & Submission	Complete APK, GitHub repository, PDF documentation, screenshots, database structure, and app logo	10 pts
TOTAL		100 pts